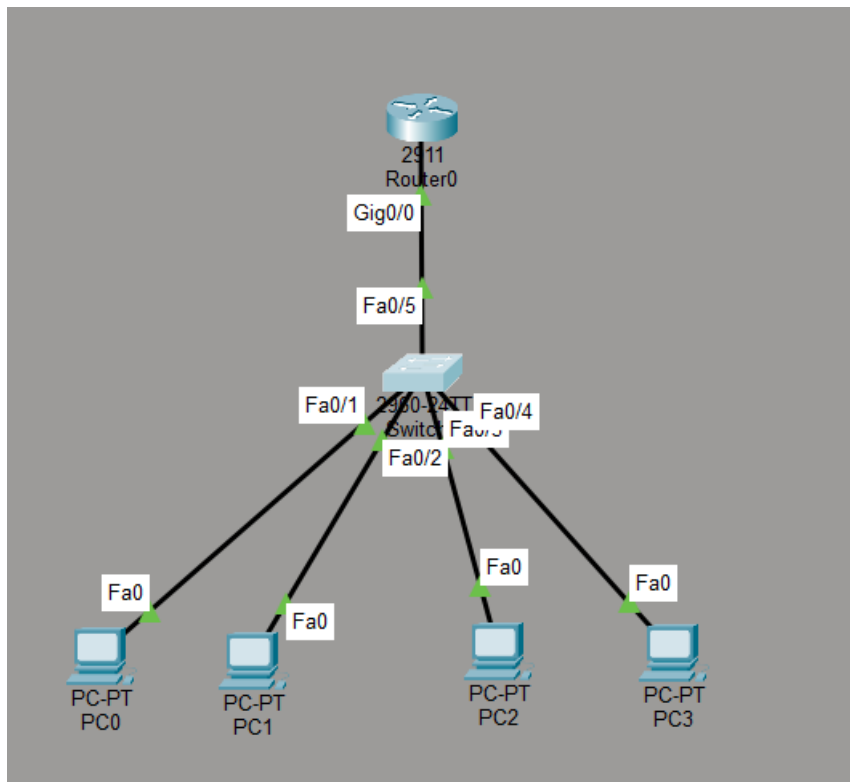


Q-Capture and Analyze Inter-VLAN Traffic Use Wireshark to observe how frames change as they cross from switch to router and back. (Integrative — multi-step flow combining VLANs, trunking, and routing.) Capture ICMP traffic while pinging between a VLAN 10 host and a VLAN 20 host. Identify the specific frames that cross the router via the trunk link. Document the source/destination MAC and IP addresses observed at each hop. Explain how the frame headers change as traffic passes from switch to router and back.

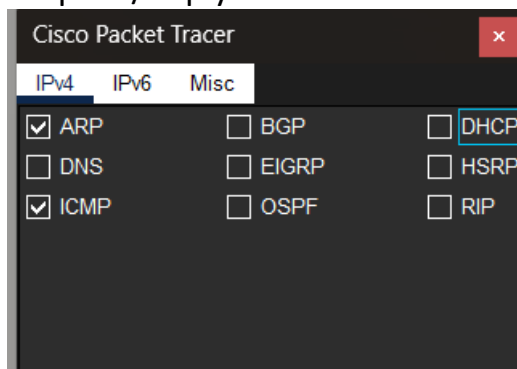
To capture and analyze ICMP traffic between a host in VLAN 10 and a host in VLAN 20, observe the traffic as it crosses the switch-router trunk, and document how Layer-2 (MAC/VLAN) headers change while the Layer-3 IP addresses remain end-to-end.

Topology :



Simulation Filter Configuration

Packet Tracer Simulation Mode was used. ARP and ICMP were selected so that the ICMP Echo Request/Reply traffic could be isolated from unrelated protocols.



ICMP Capture

After resetting the simulation, an ICMP ping was generated from PC0 to PC2. The Event List shows the ICMP packet moving from PC0 to Switch0, then to Router0, back through Switch0, and finally to PC2. The reverse path carries the ICMP Echo Reply.

Simulation Panel

Event List

Vis.	Time(sec)	Last Device	At Device	Type
—	2.869	Switch0	Router0	STP
—	2.869	Switch0	PC0	STP
—	3.263	--	PC0	ICMP
—	3.264	PC0	Switch0	ICMP
—	3.265	Switch0	Router0	ICMP
—	3.266	Router0	Switch0	ICMP
—	3.267	Switch0	PC2	ICMP
—	3.268	PC2	Switch0	ICMP
—	3.269	Switch0	Router0	ICMP
—	3.270	Router0	Switch0	ICMP
—	3.271	Switch0	PC0	ICMP
—	3.601	--	Switch0	STP
—	3.602	Switch0	Router0	STP
—	3.602	Switch0	PC3	STP
—	3.602	--	Switch0	STP
—	3.603	Switch0	Router0	STP
—	4.000	--	Switch0	STP
—	4.001	Switch0	Router0	STP
—	4.001	Switch0	PC2	STP
—	4.001	Switch0	PC1	STP

Reset Simulation

☒ Constant Delay

Captured to 12.000 s

Play Controls

⏮

⏸

⏭

Frame Analysis at the Switch

The Switch0 PDU information shows the Ethernet frame entering the switch and the outgoing frame carrying a Dot1Q header toward the router. This confirms that the switch-router connection is carrying tagged VLAN traffic.

PDU Information at Device: Switch0

OSI Model

Inbound PDU Details

Outbound PDU Details

Layer7

Layer6

Layer5

Layer4

Layer3

Layer 2: Ethernet II Header
0040.0B8C.7355 >> 0001.C953.5001

Layer 1: Port FastEthernet0/4

Layer7

Layer6

Layer5

Layer4

Layer3

Layer 2: Dot1q Header 0040.0B8C.7355
>> 0001.C953.5001

Layer 1: Port(s): FastEthernet0/1

1. FastEthernet0/4 receives the frame.

Challenge Me

<< Previous Layer

Next Layer >>

Event List Filters - Visible Events

Frame Analysis at Router0

At Router0, the inbound ICMP packet is shown with source IP 192.168.10.2 and destination IP 192.168.20.2. The ICMP message is Type 8, which is an Echo Request. The Layer-2 information is tagged with 802.1Q.

PDU Information at Device: Router0

OSI Model

Inbound PDU Details

Outbound PDU Details

Layer7

Layer6

Layer5

Layer4

Layer 3: IP Header Src. IP:
192.168.10.2, Dest. IP: 192.168.20.2
ICMP Message Type: 8

Layer 2: Dot1q Header
00D0.FFBC.9E7D >> 0001.C953.5001

Layer 1: Port GigabitEthernet0/0

Layer7

Layer6

Layer5

Layer4

Layer 3: IP Header Src. IP:
192.168.10.2, Dest. IP: 192.168.20.2
ICMP Message Type: 8

Layer 2: Dot1q Header 0001.C953.5001
>> 0040.0B8C.7355

Layer 1: Port(s): GigabitEthernet0/0

1. GigabitEthernet0/0 receives the frame.

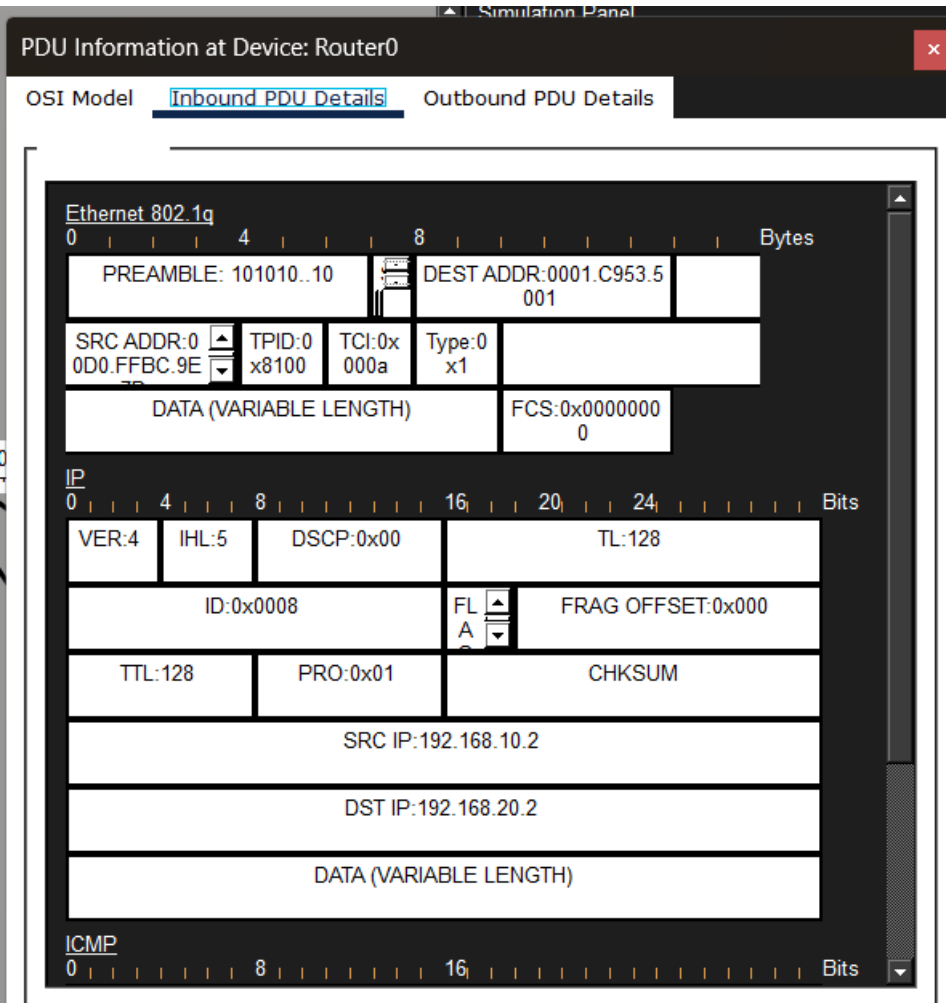
Challenge Me

<< Previous Layer

Next Layer >>

Detailed Inbound 802.1Q and IP Header

The detailed inbound PDU shows TCI = 0x000A. Hexadecimal 0x000A equals decimal 10, therefore the incoming frame is tagged as VLAN 10. The same PDU shows the ICMP packet with source IP 192.168.10.2 and destination IP 192.168.20.2.



Observed Header Values

The following values are taken directly from the captured Packet Tracer PDU information.

Hop / Frame	Source MAC	Destination MAC	Source IP	Destination IP	VLAN / Tag
Switch0 → Router0 (observed)	0001.C953.9E7D	0001.C953.5001	192.168.10.2	192.168.20.2	VLAN 10 (TCI 0x000A)
Router0 outbound (observed)	0001.C953.5001	0040.0B8C.7355	192.168.10.2	192.168.20.2	Dot1Q shown; outbound VLAN tag value not visible in captured screenshot

How the Frame Headers Change

When the ICMP packet travels from VLAN 10 toward VLAN 20, the IP packet is routed by Router0. The Layer-2 Ethernet information is therefore rebuilt for the next Layer-2 segment. The captured values demonstrate that the MAC addresses change across the routed hop, while the source and destination IP addresses remain 192.168.10.2 and 192.168.20.2.

Before / inbound to Router0	MAC: 0001.C953.9E7D → 0001.C953.5001 IP: 192.168.10.2 → 192.168.20.2 VLAN tag: 10
After / outbound from Router0	MAC: 0001.C953.5001 → 0040.0B8C.7355 IP: 192.168.10.2 → 192.168.20.2 New Layer-2 frame is created
Main observation	MAC addresses change at the routed hop; IP source/destination remain end-to-end for this ICMP exchange.

The Inter-VLAN ICMP traffic was successfully captured in Cisco Packet Tracer Simulation Mode. The capture shows an 802.1Q-tagged frame entering Router0 on VLAN 10, carrying an ICMP Echo Request from 192.168.10.2 to 192.168.20.2. Router0 routes the packet toward the destination network and creates a new Layer-2 frame, changing the MAC addresses while preserving the end-to-end IP addresses. This demonstrates the interaction of VLANs, trunking, Layer-2 switching, and Layer-3 routing.